From Heuristic Trajectory Scanning to Spatial Language Modeling: Graph-Constrained Trajectory Transformers for Dynamic Arterial Soft Route Discovery in Large-Scale Microscopic AMoD Simulations

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Abstract

This research document establishes the theoretical formulation and mathematical architecture for Version 4.0 of the Last-Mile Commute and Autonomous Mobility-on-Demand (AMoD) framework within the Manhattan SUMO simulation ecosystem. In prior versions (V2.0 and V3.0), predictive dispatch and spatial decomposition relied on "soft routes" (arterial corridors) generated via TopoScan, a heuristic trajectory clustering algorithm. While functional, TopoScan suffers from fundamental structural vulnerabilities: brittle exact-sequence matching, artificial Dijkstra shortest-path bias, inability to generalize across temporal windows, and total collapse during sparse off-peak periods.

To definitively solve these limitations without empirical ad-hoc tuning, we reframe arterial corridor discovery as a **Spatial-Temporal Natural Language Processing (NLP) sequence generation problem**. Road network edges are formalized as discrete lexical tokens, vehicle trajectories as grammatical sentences, and dynamic soft routes as high-probability semantic phrases (idioms) conditioned on temporal context. We propose the **Graph-Constrained Trajectory Language Model (GCT-LM)**, an autoregressive Causal Transformer equipped with continuous temporal embeddings and a **Topological Adjacency Mask** derived directly from the Manhattan road network graph. This mask mathematically guarantees that generated corridors are 100% physically connected and routable within SUMO, eliminating hallucinations. We detail the complete mathematical formulation, define every variable and operator, map the analytical advantages over heuristic scanning, and integrate the learned corridors into the downstream Game-Theoretic V3 dispatch engine (Urn-Ball matching, Pigouvian subsidies, and Virtual Parking).

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1 Introduction and System Context

1.1 The Simulation Ecosystem: SUMO-LMC

The simulation framework under study models an Autonomous Mobility-on-Demand (AMoD) fleet servicing thousands of concurrent on-demand passenger requests across the Manhattan street network. The environment is grounded in physical microscopic simulation via the Simulation of Urban MObility (SUMO) engine, orchestrated through TraCI and `libsumo` interfaces using real-world demand derived from the New York City Yellow Taxi dataset.

At metropolitan scale (e.g., 900 to 3,000 active vehicles and over 3,000 requests per hour), dial-a-ride ridesharing faces an exponential computational bottleneck. Evaluating all possible groupings of k passengers across V available vehicles scales factorially:

$$\mathcal{O}\left(V \cdot \binom{P}{k} \cdot (2k)! \cdot c^{2k}\right) \tag{1}$$

where P is the unassigned passenger pool and c represents the discrete pickup/dropoff stop alternatives per passenger. A monolithic, centralized combinatorial search instantly saturates computing resources and triggers real-time simulation deadlocks.

1.2 The Role of Soft Routes

To overcome this combinatorial explosion, Versions 2.0 and 3.0 introduced **Hierarchical Spatial Decomposition** driven by **Soft Routes** (also referred to as *trunks* or *corridors*).

A soft route is a directed, high-density backbone path along major urban arteries (e.g., Broadway, 5th Avenue, FDR Drive). Rather than operating as rigid fixed transit lines (like light rail or buses), soft routes serve as flexible spatial guides:

(a) **Search Space Decoupling (Bucketing):** Vehicles and passengers are partitioned into independent spatial subproblems based on proximity to active trunks. Combinatorial subset optimization is executed in parallel across isolated threads without cross-route interference.

(b) **Dynamic Pooling via Willingness-to-Walk (WtW):** Instead of requiring vehicles to execute costly door-to-door zig-zag detours through congested cross-streets, passengers within a spatial catchment radius (e.g., 500 meters) walk to the trunk entry access point and walk from the egress point, concentrating passenger flow onto high-capacity arteries.

(c) **Predictive Fleet Equilibrium:** By forecasting trunk demand 30 to 60 minutes into the future, the platform proactively guides idle vehicles toward stressed corridors using Pigouvian subsidies and virtually parks surplus vehicles, maintaining a Wardrop User Equilibrium.

The entire operational stability and pooling throughput of the simulation depends directly on the **quality, continuity, representativeness, and temporal responsiveness of these soft routes**.

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2 Problem Formulation: Structural Deficiencies of TopoScan

In the current codebase (`train_soft_routes.py`), soft routes are generated using **TopoScan**, a trajectory clustering heuristic. TopoScan segments the 24-hour day into 48 overlapping 30-minute bins, samples trip origin-destination pairs, routes them via Dijkstra's algorithm, and attempts to grow arterial trunks from the most traversed edges.

Comprehensive theoretical and code-level audits reveal four critical structural flaws that compromise the scientific validity and operational efficiency of TopoScan:

2.1 Brittle Exact Sequential Subpath Matching

TopoScan iteratively expands a candidate trunk $\text{TK} = [e_1, e_2, \ldots, e_m]$ by checking whether the subpath is strictly contained within historical paths:

$$\mathcal{C}(\text{TK}) = \{p \in \mathcal{P} \mid \text{TK} \sqsubseteq p\} \tag{2}$$

Extension terminates the instant $|\mathcal{C}(\text{TK})| < \alpha$ (where $\alpha = 5$ is a hardcoded cluster threshold).

In a complex urban grid like Manhattan, taxi trajectories exhibit high local variance. Drivers frequently alternate between parallel avenues (e.g., 5th Avenue vs. Madison Avenue) or execute one-block diversion maneuvers around localized congestion, double-parked delivery vehicles, or traffic signals. Because TopoScan enforces strict contiguous sequence equality ($\sqsubseteq$), a single edge deviation causes an otherwise identical corridor trip to be dropped from $\mathcal{C}(\text{TK})$. This causes:

- **Premature Trunk Truncation:** Corridors terminate abruptly mid-avenue because flow temporarily splits across parallel lanes or adjacent streets.
- **Corridor Fragmentation:** Multiple disconnected 2-edge or 3-edge fragments are generated instead of cohesive, continuous cross-borough arterials.

2.2 Synthetic Shortest-Path Bias (Dijkstra Artifacts)

In `train_soft_routes.py`, TopoScan does not cluster actual GPS trajectory paths; instead, it takes raw pickup/dropoff coordinates, snaps them to spatial hubs, and synthesizes paths using Dijkstra's shortest path algorithm:

$$p^* = \arg \min_{p \in \text{Paths}(u,v)} \sum_{e \in p} \text{Length}(e) \tag{3}$$

This introduces severe methodological bias:

- Dijkstra routes vehicles based purely on static Euclidean link lengths, ignoring traffic light cycles, speed limits, multi-lane capacities, turn penalties, and empirical congestion.
- In real Manhattan traffic, drivers rarely take static shortest-distance paths; they utilize dynamic arterial bypasses (e.g., West Side Highway, FDR Drive) to maintain speed. TopoScan is blind to these latent empirical flows.

2.3 Temporal Discontinuity and Low-Demand Collapse

TopoScan operates on each of the 48 time bins in complete mathematical isolation. Bin b shares zero parameters, representations, or statistics with Bin $b-1$ or Bin $b+1$.

- During off-peak windows (e.g., 02:00 AM to 05:00 AM), trip volume drops below the empirical density threshold. TopoScan finds zero trunks ($|\text{trunks}| = 0$).
- In the simulation engine (`03_run_ridesharing.py`), when $|\text{trunks}| = 0$, the spatial decomposition layer collapses. The entire fleet is forced into the Grounded Reserve Fleet, reverting the system to an inefficient, point-to-point unpooled dial-a-ride model.
- Real traffic exhibits smooth temporal autocorrelation: the network state at 08:30 AM is a continuous evolution of 08:00 AM. TopoScan's discrete, independent binning treats every window as an isolated universe.

2.4 Lack of Global Probabilistic Sequence Modeling

TopoScan grows trunks via a greedy, local transition rule:

$$e^* = \arg\max_{e \in \mathcal{E}_{\text{cand}}} \text{Frequency}(e \mid \text{TK}) \tag{4}$$

This greedy step is inherently myopic. It possesses no global objective function, no awareness of total sequence likelihood, and no ability to quantify uncertainty. It cannot evaluate whether a corridor is representative of 80% of travelers or merely an accidental artifact of a few clustered trips.

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3 The NLP Paradigm Shift: Trajectories as Spatial Language

3.1 The Formal Isomorphism

To resolve these deficiencies without ad-hoc heuristics, we leverage the fundamental mathematical isomorphism between discrete Natural Language Processing and road-network trajectory dynamics.

Under this formulation, identifying a soft route is no longer an arbitrary geometric clustering problem. Instead, it is rigorously defined as:

> *Finding the Maximum A Posteriori (MAP) sequence of spatial tokens generated by a probabilistic language model conditioned on a specific temporal context, subject to physical grammatical (topological) constraints.*

3.2 Vocabulary Definition over the LSCC

Let $G = (V, E)$ represent the directed road network graph of Manhattan extracted from OpenStreetMap and compiled into SUMO. To ensure all generated routes are mathematically and physically viable, we restrict the vocabulary $\mathcal{V}$ to the **Largest Strongly**

Table 1: Formal Isomorphism: Language Modeling vs. Trajectory Corridor Discovery

NLP Concept	**Urban Mobility Counterpart**	**SUMO / Simulation Implementation**
Lexicon / Vocabulary ($\mathcal{V}$)	Road network directed edges	Drivable edges in Manhattan LSCC network
Token ($w_t \in \mathcal{V}$)	Discrete road segment (e_t)	SUMO Edge ID (e.g., `"4284910#0"`)
Sentence ($\mathbf{w} = [w_1, \ldots, w_n]$)	Trajectory / Trip ($\mathbf{e} = [e_1, \ldots, e_n]$)	Sequential edge path traversed by vehicle
Grammar & Syntax Rules	Physical network topology	Legal outgoing turns, one-way constraints
Semantic Corpus ($\mathcal{D}$)	Historical mobility dataset	NYC Yellow Taxi trip records (7 days)
Frequent Idiom / N-gram	Arterial Soft Route (Trunk)	Cohesive high-capacity transport corridor
Temporal Context / Prompt	Time of day / Day of week	Discrete 30-min bin identifier $b \in \{0, \ldots, 47\}$

Connected Component (LSCC) of the drivable network:

$$\mathcal{V} = \{e \in E \mid e \text{ allows passenger vehicles and } e \in \text{LSCC}(G)\} \tag{5}$$

Let $|\mathcal{V}| = N_e$ denote the vocabulary size (approximately 4,500 to 6,000 distinct directed edge segments in Manhattan). Every edge is assigned a unique index $i \in \{1, 2, \ldots, N_e\}$.

We add special tokens standard in sequence modeling:

- [PAD]: Index 0, used for batch padding.
- [BOS]: Beginning-of-sequence token.
- [EOS]: End-of-sequence token (trip destination reached).
- [BIN_b]: Temporal prefix tokens for $b \in \{0, 1, \ldots, 47\}$.

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4 The Proposed Model: Graph-Constrained Trajectory Language Model (GCT-LM)

4.1 Input Representation and Multi-Modal Embedding Layer

A trip trajectory is represented as a sequence of edge tokens $\mathbf{e} = [e_1, e_2, \ldots, e_L]$ occurring within a discrete 30-minute time bin $b \in \{0, \ldots, 47\}$.

For each token at sequence position $t \in \{1, \ldots, L\}$, the input embedding $\mathbf{h}_t^{(0)} \in \mathbb{R}^d$ (where d is the hidden dimension, typically $d = 256$) is constructed as the additive fusion of three dedicated embedding manifolds:

$$\mathbf{h}_t^{(0)} = \mathbf{E}_{\text{edge}}(e_t) + \mathbf{E}_{\text{time}}(b) + \mathbf{E}_{\text{pos}}(t) \tag{6}$$

4.1.1 Variable Definitions and Mechanics

1. **Edge Token Embedding Matrix $\mathbf{E}_{\text{edge}} \in \mathbb{R}^{(|\mathcal{V}|+K)\times d}$:** Maps discrete road segment IDs to continuous semantic feature vectors. Spatial proximity and co-occurrence patterns are naturally captured: edges along the same avenue will project closely in the embedding manifold through gradient descent.

2. **Continuous Temporal Embedding $\mathbf{E}_{\text{time}} \in \mathbb{R}^{48\times d}$:** Embeds the 30-minute temporal bin b. Unlike TopoScan's disjoint bins, temporal embeddings are regularized to ensure temporal smoothness. Let $t_{\text{sec}} = b \times 1800$. The initial temporal representation uses continuous harmonic encodings:

$$\mathbf{E}^{(0)}_{\text{time}}(b)_{2i} = \sin\left(\frac{2\pi \cdot b \cdot i}{48 \cdot d}\right), \quad \mathbf{E}^{(0)}_{\text{time}}(b)_{2i+1} = \cos\left(\frac{2\pi \cdot b \cdot i}{48 \cdot d}\right) \tag{7}$$

   During training, adjacent bins share gradient information through continuous attention, allowing low-demand off-peak bins to borrow structural statistical strength from adjacent periods.

3. **Learnable Positional Encoding $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{L_{\max}\times d}$:** Encodes the sequential order of road segments within the trip, preserving the physical directional orientation of traffic flow.

4.2 Autoregressive Transformer Backbone

The sequence of input representations $\mathbf{H}^{(0)} = [\mathbf{h}^{(0)}_1, \mathbf{h}^{(0)}_2, \ldots, \mathbf{h}^{(0)}_L] \in \mathbb{R}^{L\times d}$ passes through a stack of N_{layers} causal Transformer blocks ($N_{\text{layers}} \in [4, 6]$, multi-head attention heads $H = 8$).

For layer $l \in \{1, \ldots, N_{\text{layers}}\}$:

$$\mathbf{Q}^{(l)} = \mathbf{H}^{(l-1)}\mathbf{W}^{(l)}_Q, \quad \mathbf{K}^{(l)} = \mathbf{H}^{(l-1)}\mathbf{W}^{(l)}_K, \quad \mathbf{V}^{(l)} = \mathbf{H}^{(l-1)}\mathbf{W}^{(l)}_V \tag{8}$$

$$\mathbf{A}^{(l)}_{\text{causal}} = \text{Softmax}\left(\frac{\mathbf{Q}^{(l)}(\mathbf{K}^{(l)})^T}{\sqrt{d_k}} + \mathbf{M}_{\text{causal}}\right)\mathbf{V}^{(l)} \tag{9}$$

$$\mathbf{H}'^{(l)} = \text{LayerNorm}\left(\mathbf{H}^{(l-1)} + \mathbf{A}^{(l)}_{\text{causal}}\right) \tag{10}$$

$$\mathbf{H}^{(l)} = \text{LayerNorm}\left(\mathbf{H}'^{(l)} + \text{FFN}(\mathbf{H}'^{(l)})\right) \tag{11}$$

where $\mathbf{M}_{\text{causal}} \in \mathbb{R}^{L\times L}$ is the standard upper-triangular autoregressive mask:

$$(\mathbf{M}_{\text{causal}})_{i,j} = \begin{cases} 0 & \text{if } i \geq j \\ -\infty & \text{if } i < j \end{cases} \tag{12}$$

ensuring that the prediction of edge e_{t+1} depends strictly on preceding edges $[e_1, \ldots, e_t]$.

The final hidden representation $\mathbf{h}^{(N)}_t \in \mathbb{R}^d$ is projected onto the road vocabulary via the output projection head:

$$\mathbf{z}_t = \mathbf{W}_{\text{vocab}}\mathbf{h}^{(N)}_t + \mathbf{b}_{\text{vocab}} \quad \in \mathbb{R}^{|\mathcal{V}|} \tag{13}$$

where $\mathbf{z}_t$ represents the unconstrained raw logit scores for all edges in the Manhattan network.

4.3 Zero-Hallucination Topological Adjacency Masking

In generic language modeling, a model is free to output any word from the vocabulary. In traffic simulation, however, an autonomous vehicle cannot teleport; it can **only drive onto an edge that physically connects to its current edge**.

If an unconstrained Transformer were used, it would inevitably produce disconnected road sequences (hallucinations), causing TraCI to fail with `RouteReplacementError` during simulation runtime.

To solve this definitively, we introduce a **Topological Adjacency Mask** directly into the softmax probability layer.

Let $\mathcal{N}^+(e_t)$ denote the set of physically valid outgoing adjacent edges from edge e_t in the SUMO network:

$$\mathcal{N}^+(e_t) = \{e' \in \mathcal{V} \mid \text{FromNode}(e') = \text{ToNode}(e_t)\} \tag{14}$$

We define the state-dependent graph transition mask $\mathbf{M}_{\text{topo}}(e_t) \in \mathbb{R}^{|\mathcal{V}|}$ as:

$$(\mathbf{M}_{\text{topo}}(e_t))_j = \begin{cases} 0 & \text{if } j \in \mathcal{N}^+(e_t) \\ -\infty & \text{if } j \notin \mathcal{N}^+(e_t) \end{cases} \tag{15}$$

The topologically constrained next-edge probability distribution is computed as:

$$P(e_{t+1} = j \mid e_{\leq t}, b) = \frac{\exp\left((\mathbf{z}_t)_j + (\mathbf{M}_{\text{topo}}(e_t))_j\right)}{\sum_{k \in \mathcal{V}} \exp\left((\mathbf{z}_t)_k + (\mathbf{M}_{\text{topo}}(e_t))_k\right)} \tag{16}$$

4.3.1 Mathematical Guarantee

Because $\exp(-\infty) = 0$, for all edges $j \notin \mathcal{N}^+(e_t)$, the assigned probability is identically zero:

$$\forall j \notin \mathcal{N}^+(e_t), \quad P(e_{t+1} = j \mid e_{\leq t}, b) \equiv 0 \tag{17}$$

This mathematical constraint **guarantees with 100% certainty that every generated sequence is continuous, topologically contiguous, and physically drivable within the SUMO microscopic simulator**.

4.4 Self-Supervised Training Objective

The model is trained end-to-end on historical trip sequences using the Next-Token Prediction (NTP) negative log-likelihood objective:

$$\mathcal{L}_{\text{NTP}}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(\mathbf{e}, b) \in \mathcal{D}} \sum_{t=1}^{L-1} \log P_\theta(e_{t+1} \mid e_1, \ldots, e_t, b) \tag{18}$$

Through backpropagation, the model learns the complex, non-linear conditional transition probabilities of urban traffic across time, capturing which avenues are favored, which turns are avoided, and how commuting demand evolves hour by hour.

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5 Corridor Extraction via Graph-Constrained Beam Search

Once the GCT-LM is trained, extracting the optimal soft routes for time bin b transitions from an ad-hoc heuristic scan into a principled **Maximum A Posteriori (MAP) decoding problem**.

5.1 The MAP Corridor Formulation

For a given time bin b and a specified candidate seed entry edge $e_1 \in \mathcal{V}_{\text{entry}}$, we seek the continuous edge sequence $\text{Trunk}^* = [e_1, e_2, \ldots, e_m]$ that maximizes the cumulative conditional log-likelihood subject to a minimum length constraint:

$$\text{Trunk}^* = \arg\max_{e_1,\ldots,e_m} \sum_{t=1}^{m-1} \log P(e_{t+1} \mid e_{\leq t}, b) \tag{19}$$

$$\text{subject to:} \quad e_{t+1} \in \mathcal{N}^+(e_t) \quad \forall t \in \{1, \ldots, m-1\} \tag{20}$$

$$\sum_{t=1}^{m} \text{Length}(e_t) \geq L_{\min} \quad (L_{\min} = 1.5 \text{ km}) \tag{21}$$

$$e_i \neq e_j \quad \forall i \neq j \quad \text{(Loop-free constraint)} \tag{22}$$

5.2 Graph-Constrained Beam Search Decoding

Because evaluating all legal paths across the network is computationally prohibitive, we employ **Constrained Beam Search** with beam width W (e.g., $W = 5$ to 10):

1. **Seed Identification:** Identify the top K high-volume origin edges in bin b (e.g., major entry gateways: Lincoln Tunnel access, FDR Drive entries, Financial District arterials).

2. **Beam Expansion:** At step t, expand each active hypothesis in the beam only across its topologically legal outgoing edges $\mathcal{N}^+(e_t)$.

3. **Pruning:** Retain the top W candidate sequences based on normalized cumulative log-likelihood:

$$\text{Score}(\text{TK}) = \frac{1}{|\text{TK}|^\gamma} \sum_{t=1}^{|\text{TK}|-1} \log P(e_{t+1} \mid e_{\leq t}, b) \tag{23}$$

   where $\gamma \in [0.6, 0.8]$ is a length penalty preventing premature termination.

4. **Redundancy Filtering:** To avoid generating near-identical duplicate trunks, compute pairwise Jaccard edge overlap:

$$J(\text{TK}_a, \text{TK}_b) = \frac{|\text{TK}_a \cap \text{TK}_b|}{|\text{TK}_a \cup \text{TK}_b|} \tag{24}$$

   If $J(\text{TK}_a, \text{TK}_b) > 0.70$, retain only the higher-scoring trunk.

5.3 Complementary Non-Neural Baseline: Spatial Byte-Pair Encoding (Spatial-BPE)

As an immediate, interpretable counterpart, the NLP tokenization algorithm **Byte-Pair Encoding (BPE)** can also be adapted to spatial sequences without neural network training:

- Start with individual edges as base tokens: $\mathcal{V}_0 = E$.
- Count the frequency of all adjacent edge pairs (e_i, e_j) across all historical trips in bin b, where $\text{ToNode}(e_i) = \text{FromNode}(e_j)$.
- Iteratively merge the most frequent adjacent pair into a new composite super-token: $e_{\text{new}} = (e_i \circ e_j)$.
- Terminate when the length of the merged super-tokens exceeds $L_{\min} = 1.5$ km or frequency drops below threshold.

This provides a fast, statistical subword baseline that inherently preserves graph connectivity while avoiding TopoScan's brittle subpath matching.

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6 Integration with the Downstream Game-Theoretic Simulation Engine

A critical requirement of the Version 4.0 architecture is that the learned soft routes must integrate seamlessly into the existing `03_run_ridesharing.py` engine without altering its downstream Game-Theoretic mathematics.

The output of GCT-LM is formatted identically to the schema required by the simulator:

$$\texttt{dynamic_soft_routes.json} = \{b \in \{0, \ldots, 47\} : [\text{Trunk}_1, \text{Trunk}_2, \ldots, \text{Trunk}_K]\} \quad (25)$$

where each trunk is a serialized list of SUMO edge strings (e.g., `["4284910#0", "4284910#1", ...]`).

Once loaded into the simulation, these corridors govern three core operational layers:

6.1 1. Spatial Catchment Buffering (Willingness to Walk)

For active soft route i with entry edge $Access_i = \text{Trunk}_i[0]$ and exit edge $Egress_i = \text{Trunk}_i[-1]$, passenger p is triaged into bucket i if their origin and destination lie within catchment radius $R_{\text{catchment}} = 500$m:

$$M_{p,i} = \begin{cases} \text{Dist}(O_p, Access_i) + \text{Dist}(D_p, Egress_i) & \text{if } \|O_p - Access_i\| \leq R \text{ and } \|D_p - Egress_i\| \leq R \\ \infty & \text{otherwise} \end{cases} \quad (26)$$

Because GCT-LM corridors represent true high-volume traffic flows, passenger triage success rates will be substantially higher than TopoScan's synthetic Dijkstra corridors.

6.2 2. Market Disequilibrium Index (Ω_i) and Urn-Ball Matching

For each trunk i, the simulator continuously monitors passenger demand D_i and active vehicle supply S_i to calculate network disequilibrium:

$$\Omega_i = \frac{D_i - S_i}{D_i + S_i} \in [-1, 1] \tag{27}$$

Under Cobb-Douglas / Urn-Ball matching, the probability of an idle vehicle securing a passenger match upon traversing trunk i is:

$$P(\text{match}_i) = 1 - \exp\left(-\frac{D_i}{S_i + 1}\right) \tag{28}$$

6.3 3. Pigouvian Subsidies and Virtual Parking

The platform resolves the Principal-Agent dilemma by evaluating the net economic cost for driver v servicing subset S:

$$\text{Cost}_{\text{final}} = (\gamma \cdot C_{\text{op}} \cdot \Delta_{\text{new}}) + \left(\alpha \sum_{\text{pax}} \left(1 - e^{-\beta \cdot \text{WalkDist}}\right)\right) - B(\Omega_i) \tag{29}$$

where $B(\Omega_i)$ is the Pigouvian subsidy rewarding drivers who navigate to stressed routes ($\Omega_i > 0$):

$$B(\Omega_i) = \text{Base} \times \max(0, \Omega_i)^{\text{expo}} \tag{30}$$

Conversely, when $\Omega_i < 0$ (oversupply), surplus vehicles are assigned to **Virtual Parking**, preventing empty fleet cruising from creating endogenous congestion.

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7 Analytical Improvements: Theoretical and Structural Analysis

Because this document represents the design and theoretical justification prior to simulation execution, we assess the proposed architecture through **formal analytical and structural comparisons** against TopoScan:

7.1 Detailed Analytical Impacts

7.1.1 1. Resilience to Grid Divergence and Parallel Avenues

Under TopoScan, two taxi trajectories that share 90% of their route along 5th Avenue but diverge for two blocks onto Madison Avenue are treated as disjoint sets. In GCT-LM, the self-attention mechanism computes attention weights across long contexts:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{31}$$

The model naturally learns that 5th Avenue and Madison Avenue share parallel semantic roles in northbound flow. As a result, the beam search extracts robust, representative backbone trunks that capture the full volume of both flows rather than prematurely terminating.

Table 2: Analytical Comparison: TopoScan vs. GCT-LM Architecture

Analytical Dimension	**Current Heuristic (TopoScan)**	**Proposed GCT-LM (NLP)**
Sequence Matching	Brittle discrete subpath ($\sqsubseteq$)	Probabilistic continuous likelihood $P(\mathbf{e} \mid b)$
Grid Robustness	Dropped on 1-block detour	Generalizes across parallel avenues
Data Source	Synthetic Dijkstra shortest paths	Empirical historical taxi trajectory flows
Temporal Regularity	Independent, disconnected bins	Shared continuous temporal manifold $\mathbf{E}_{\text{time}}(b)$
Off-Peak Stability	Complete collapse ($\lvert\text{trunks}\rvert = 0$)	Smooth interpolation from adjacent hours
Routability Guarantee	Heuristic (can produce dead-ends)	100% mathematically guaranteed by $\mathbf{M}_{\text{topo}}$
Objective Function	None (greedy local edge extension)	Global MAP sequence optimization
Corridor Redundancy	Highly overlapping duplicated paths	Formal Jaccard diversity constraint

7.1.2 2. Differentiable Temporal Continuity

In TopoScan, if Bin 7 (03:30 AM) contains only 120 trips, the heuristic fails to find clusters above $\alpha = 5$. In GCT-LM, the temporal embedding $\mathbf{E}_{\text{time}}(b)$ ensures that:

$$\|\mathbf{E}_{\text{time}}(b) - \mathbf{E}_{\text{time}}(b \pm 1)\| < \epsilon \tag{32}$$

The model smoothly interpolates arterial corridors during low-volume periods by borrowing structural representations from higher-density boundary hours (e.g., 02:00 AM and 06:00 AM). This eliminates the complete collapse of soft routes during off-peak hours.

7.1.3 3. Zero Microscopic Routing Deadlocks

In SUMO, calling `traci.vehicle.setRoute(vid, edge_list)` when consecutive edges do not form a physically connected graph component triggers an unrecoverable C++ exception. While TopoScan occasionally produces disconnected fragments due to OSM hub snapping artifacts, GCT-LM's topological mask $\mathbf{M}_{\text{topo}}$ forces non-adjacent transition probabilities to zero. Consequently, routing failure rates in TraCI drop to **identically zero**.

7.1.4 4. Downstream Ride-Pooling Impact

When soft routes reflect actual empirical traffic flows rather than synthetic Dijkstra paths:

- More passenger origin-destination pairs naturally fall within the 500m spatial catchment buffer of trunk access/egress points.
- The proportion of passengers assigned to high-capacity pooled trunks increases, reducing reliance on the unpooled Reserve Fleet.

- Physical vehicle occupancy increases while passenger detour delays (Δ_{new}) decrease, because vehicles are traversing natural travel corridors rather than forced geometric detours.

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8 Conclusion and Next Steps

This document has established the theoretical foundations of the **Graph-Constrained Trajectory Language Model (GCT-LM)** for soft route generation in AMoD simulations. By replacing the heuristic TopoScan algorithm with a mathematically grounded sequence model, the architecture:

1. Casts trajectory corridor mining into a rigorous probabilistic language modeling framework.
2. Guarantees 100% physical network continuity and drivability through topological adjacency masking.
3. Resolves off-peak temporal collapse via continuous temporal embeddings.
4. Provides drop-in compatibility with the existing V3 Game-Theoretic SUMO dispatch framework.

Following architectural review, future work will proceed to offline model training on the 7-day Manhattan dataset, export to `dynamic_soft_routes.json`, and empirical validation through full-scale SUMO simulation runs evaluated via `DataAnalysis_V3.py`.

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